

Your grade: 100%

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Next item →

1.

What are the two main categories of parallel computing problems mentioned in the lecture?

1 / 1 point

☒ Coupled parallel and ideally parallel

☐ Shared memory and distributed memory

☐ Strong scaling and weak scaling

☐ Single Program Multiple Data (SPMD) and Multiple Program Multiple Data (MPMD)

 Nice work

Correct, these are the two categories mentioned in the lecture.
2.

Which of the following is an example of a coupled parallel application?

1 / 1 point

☐ Rendering different frames of an animation in parallel

☐ Analyzing multiple stock trading simulations simultaneously

☒ Simulating the spread of a disease across a population

☐ Scanning a network for open ports on multiple machines

 Nice work

Correct, this requires modeling the interactions and spread between individuals, which is a coupled problem.
3.

What is the definition of speedup in parallel computing?

1 / 1 point

☐ The ratio of serial runtime to parallel runtime

☐ The ratio of parallel runtime to serial runtime

☐ The difference between serial runtime and parallel runtime

☐ The product of serial runtime and parallel runtime

 Nice work

Correct, this is the definition of speedup mentioned in the lecture.
4.

In the context of parallel computing efficiency, what does an efficiency of 1 indicate?

1 / 1 point

☒ The program scales linearly with the number of cores

☐ The program is running slower than expected

☐ The program is running faster than expected

☐ The program is not utilizing the available resources efficiently

 Nice work

Correct, an efficiency of 1 means the program scales linearly.
5.

What is the main purpose of the Message Passing Interface (MPI) standard?

1 / 1 point

☐ To standardize message-passing implementations for parallel computing

☐ To provide a library for shared memory programming

☐ To enable communication between processors on the same node

☐ To facilitate data exchange between different programming languages

 Nice work

Correct, this is the main purpose of MPI mentioned in the lecture.
6.

What is true about variables in each process in the Single Program Multiple Data (SPMD) model?

1 / 1 point

☐ All variables are shared between processes

☐ All variables are global and accessible by all processes

☒ Each variable is local to the process defined in

☐ Variables can be selectively shared between processes

 Nice work

Correct, variables are local to each process in SPMD.
7.

What is the primary reason for using message passing for interprocessor communication in distributed memory computers?

1 / 1 point

☒ It enables remote direct access to memory between processors

☐ It simplifies the complexity of operating systems

☐ It is required by the MPI standard

☐ It is easier to program than shared memory approaches

 Nice work

Correct, this is the primary reason mentioned in the lecture.
8.

In the context of parallel computing, what is meant by "profiling" an application?

1 / 1 point

☒ Measuring the time taken by different sections of the program

☐ Optimizing the program for parallel execution

☐ Dividing the problem into smaller parts for distribution

☐ Identifying the appropriate problem size for the available resources

 Nice work

Correct, profiling involves timing different sections of the program.
9.

What is the purpose of using a barrier in MPI programs?

1 / 1 point

☒ To ensure that all MPI ranks are at the same line of code at the same time

☐ To enable communication between processors

☐ To divide the problem into smaller parts

☐ To optimize the program for parallel execution

 Nice work

Correct, barriers are used to synchronize MPI ranks.